an inverted-U curve, where too low a value for epsilon leads to poor learning due to lack of exploration, and too high a value for epsilon leads to poor learning due to lack of exploitation.

How can the agent perform so well even though it is acting randomly half the time? By setting the epsilon to be as high as possible until it degrades performance, we are utilizing an approximation to the principle of maximum entropy, or *maximum entropy learning*.

We can think of the entropy of an agent's policy as the amount of randomness it exhibits, and it turns out that maximizing entropy up until it starts to be counterproductive actually leads to better performance and generalization. If an agent can successfully achieve a goal even in the face of taking a high proportion of random actions, it must have a very robust policy that is insensitive to random transformations, so it will be able to handle more difficult environments.

10.6.2. Curriculum learning

We trained this agent only on the 5 × 5 version of this Gridworld environment so that it would have a small chance of randomly achieving the goal and receiving a reward. There are also bigger environments, including a 16 × 16 environment, which would make randomly winning extremely unlikely. An alternative to (or addition to) curiosity learning is to use a process called *curriculum learning*, which is when we train an agent on an easy variant of a problem, then retrain on a slightly harder variant, and keep retraining on harder and harder versions of the problem until the agent can successfully achieve a task that would have been too hard to start with. We could attempt to solve the 16 × 16 grid without curiosity by first training to maximum accuracy on the 5 × 5 grid, then retraining on the 6 × 6 grid, then the 8 × 8 grid, and finally the 16 × 16 grid.

10.6.3. Visualizing attention weights

We know we can successfully train a relational DQN on this somewhat difficult Gridworld task, but we could have used a much less fancy DQN to